

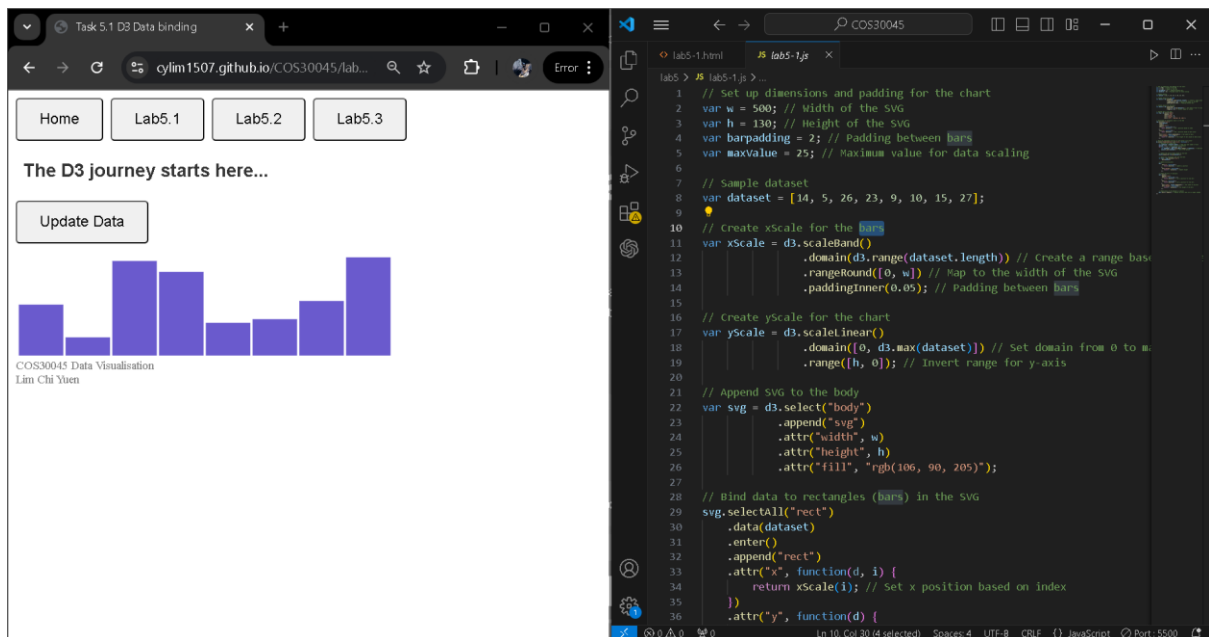
Lab Exercise Demonstration 3 (Exercises 5.1-6.2)

COS30045 Data visualisation

Lim Chi Yuen

Hosting link:

Lab5.1



Task 5.1 D3 Data binding

cyllim1507.github.io/COS30045/lab...

HomeLab5.1Lab5.2Lab5.3

The D3 journey starts here...

Update Data



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lab5-1.htmllab5-1.js

```
lab5 > lab5-1.js > ...
45 d3.select("#updateButton").on("click", function() {
60
61 // Update existing bars
62 bars
63 .attr("y", function(d) {
64   return yscale(d); // Update y position
65 })
66 .attr("height", function(d) {
67   return h - yscale(d); // Update height
68 });
69
70 // Enter new bars if necessary
71 bars.enter()
72 .append("rect")
73 .attr("x", function(d, i) {
74   return xscale(i); // Set x position for new bars
75 })
76 .attr("y", function(d) {
77   return yscale(d); // Set y position for new bars
78 })
79 .attr("width", xscale.bandwidth()) // Set width of new bars
80 .attr("height", function(d) {
81   return h - yscale(d); // Set height of new bars
82 });
83
84 // Remove bars if the dataset is smaller
85 bars.exit().remove(); // Remove any bars that are no longer needed
86 });
```

Lab5.2

Task 5.1 D3 Data binding

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Home Lab5.1 Lab5.2 Lab5.3

The D3 journey starts here...

Update Data Transition 1 Transition 2 Transition 3 Transition 4



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lab5-1.html lab5-2.html lab5-2.js

```
1 var w = 500;
2 var h = 130;
3 var barpadding = 2;
4 var maxValue = 25;
5
6 var dataset = [14, 5, 26, 23, 9, 10, 15, 27];
7
8 var xScale = d3.scaleBand()
9   .domain(d3.range(dataset.length))
10  .rangeRound([0, w])
11  .paddingInner(0.05);
12
13 var yScale = d3.scaleLinear()
14   .domain([0, d3.max(dataset)])
15   .range([h, 0]);
16
17 var svg = d3.select("body")
18   .append("svg")
19   .attr("width", w)
20   .attr("height", h)
21   .attr("fill", "rgb(106, 90, 205)");
22
23
24 svg.selectAll("rect")
25   .data(dataset)
26   .enter()
27   .append("rect")
28   .attr("x", function(d, i) {
29     return xScale(i);
30   })
31   .attr("y", function(d) {
32     return yScale(d);
33   })
34   .attr("width", xScale.bandwidth())
35   .attr("height", function(d) {
36     return h - yScale(d);
37   });
38
39
40 // Update Data Button - updates bars with new random data
41 d3.select("#updatebutton").on("click", function() {
42   // Generate new random data
43   var numValues = dataset.length;
44   dataset = [];
45   for (var i = 0; i < numValues; i++) {
46     var newNumber = Math.floor(Math.random() * maxValue);
47     dataset.push(newNumber);
48   }
49
50   // Update the yscale domain
51   yScale.domain([0, d3.max(dataset)]);
52
53   // Select all rectangles and bind new data
54   svg.selectAll("rect")
55     .data(dataset)
56     .transition()
57     .duration(2000)
58     .attr("y", function(d) {
59       return yScale(d);
60     })
61     .attr("height", function(d) {
62       return h - yScale(d);
63     });
64
65   // Transition 1 Button - example: fade out and in bars
66   d3.select("#transition3").on("click", function() {
67     svg.selectAll("rect")
```

Task 5.1 D3 Data binding

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HomeLab5.1Lab5.2Lab5.3

The D3 journey starts here...

Update DataTransition 1Transition 2Transition 3Transition 4



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lab5-1.htmllab5-2.html**lab5-2.js**

lab5 > lab5-2.js > ...

```
65 // Transition 1 Button - example: fade out and in bars
66 d3.select("#transition3").on("click", function() {
67   svg.selectAll("rect")
68     .transition()
69     .duration(1000)
70     .style("opacity", 0) // Fade out
71     .transition()
72     .duration(1000)
73     .style("opacity", 1); // Fade back in
74 });
75
76 // Transition 3 Button - oscillate width of bars
77 d3.select("#transition4").on("click", function() {
78   svg.selectAll("rect")
79     .transition()
80     .duration(1000)
81     .attr("width", xScale.bandwidth() * 1.5) // Increase width
82     .transition()
83     .duration(1000)
84     .attr("width", xScale.bandwidth()); // Restore original width
85 });
86
87 // Transition 2 Button - bars update with easing
88 d3.select("#transition1").on("click", function() {
89   // Generate new random data
90   var numValues = dataset.length;
91   dataset = [];
92   for (var i = 0; i < numValues; i++) {
93     var newNumber = Math.floor(Math.random() * maxValue);
94     dataset.push(newNumber);
95   }
96
97   // Update the yScale domain
98   yScale.domain([0, d3.max(dataset)]);
99 });
```

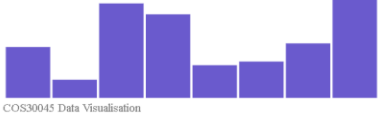
Task 5.1 D3 Data binding

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HomeLab5.1Lab5.2Lab5.3

The D3 journey starts here...

Update DataTransition 1Transition 2Transition 3Transition 4



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lab5-1.htmllab5-2.html**lab5-2.js**

lab5 > lab5-2.js > ...

```
88 d3.select("#transition1").on("click", function() {
89   // Select all rectangles and bind new data
90   svg.selectAll("rect")
91     .data(dataset)
92     .transition()
93     .delay(function(d, i){
94       return i/dataset.length * 100;
95     })
96     .duration(1000)
97     .ease(d3.easeCubicInOut) // Correct easing function
98     .attr("y", function(d) {
99       return yScale(d);
100     })
101     .attr("height", function(d) {
102       return h - yScale(d);
103     });
104 });
105
106 // Transition 2 Button - bars update with easing
107 d3.select("#transition2").on("click", function() {
108   // Generate new random data
109   var numValues = dataset.length;
110   dataset = [];
111   for (var i = 0; i < numValues; i++) {
112     var newNumber = Math.floor(Math.random() * maxValue);
113     dataset.push(newNumber);
114   }
115
116   // Update the yScale domain
117   yScale.domain([0, d3.max(dataset)]);
118
119   // Select all rectangles and bind new data
120   svg.selectAll("rect")
121     .data(dataset)
122     .transition()
123     .delay(function(d, i){
124       return i/dataset.length * 200;
125     })
126     .duration(2000)
127     .ease(d3.easeCircleIn) // Correct easing function
128     .attr("y", function(d) {
129       return yScale(d);
130     })
131     .attr("height", function(d) {
132       return h - yScale(d);
133     });
134 });
```

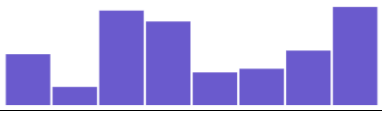
Task 5.1 D3 Data binding

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HomeLab5.1Lab5.2Lab5.3

The D3 journey starts here...

Update DataTransition 1Transition 2Transition 3Transition 4



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lab5-1.htmllab5-2.html**lab5-2.js**

lab5 > lab5-2.js > ...

```
118 d3.select("#transition2").on("click", function() {
119   // Select all rectangles and bind new data
120   svg.selectAll("rect")
121     .data(dataset)
122     .transition()
123     .delay(function(d, i){
124       return i/dataset.length * 200;
125     })
126     .duration(2000)
127     .ease(d3.easeCircleIn) // Correct easing function
128     .attr("y", function(d) {
129       return yScale(d);
130     })
131     .attr("height", function(d) {
132       return h - yScale(d);
133     });
134 });
```

Lab5.3

Task 5.1 D3 Data binding

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Home Lab5.1 Lab5.2 Lab5.3

The D3 journey starts here...

Add Remove

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lab5-1.html lab5-3.js

lab5-3.js

```
1 var w = 500;
2
3 var h = 130;
4 var barpadding = 2;
5 var maxValue = 25;
6
7 var dataset = [14, 5, 26, 23, 9, 10, 15, 27];
8
9 var xScale = d3.scaleBand()
10   .domain(d3.range(dataset.length))
11   .rangeRound([0, w])
12   .paddingInner(0.05);
13
14 var yScale = d3.scaleLinear()
15   .domain([0, d3.max(dataset)])
16   .range([h, 0]);
17
18 var svg = d3.select("body")
19   .append("svg")
20   .attr("width", w)
21   .attr("height", h)
22   .attr("fill", "rgb(106, 90, 205)");
23
24 svg.selectAll("rect")
25   .data(dataset)
26   .enter()
27   .append("rect")
28   .attr("x", function(d, i) {
29     return xScale(i);
30   })
31   .attr("y", function(d) {
32     return yScale(d);
33   })
34   .attr("width", xScale.bandwidth())
35   .attr("height", function(d) {
36     return h - yScale(d);
37   });
```

Task 5.1 D3 Data binding

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Home Lab5.1 Lab5.2 Lab5.3

The D3 journey starts here...

Add Remove

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lab5-1.html lab5-3.js

lab5-3.js

```
34
35
36   .attr("height", function(d) {
37     return h - yScale(d);
38   });
39
40 // Add new bars when "Add" button is clicked
41 d3.select("#add").on("click", function() {
42   var newNumber = Math.floor(Math.random() * maxValue);
43   dataset.push(newNumber);
44
45   // Update xScale domain with the new dataset length
46   xScale.domain(d3.range(dataset.length));
47
48   var bars = svg.selectAll("rect")
49     .data(dataset);
50
51   // Append new bars
52   bars.enter()
53     .append("rect")
54     .attr("x", w) // Start off-screen on the right
55     .attr("y", function(d) {
56       return yScale(d);
57     })
58     .attr("width", xScale.bandwidth())
59     .attr("height", function(d) {
60       return h - yScale(d);
61     })
62     .merge(bars)
63     .transition()
64     .duration(500)
65     .attr("x", function(d, i) {
66       return xScale(i);
67     })
68     .attr("y", function(d) {
69       return yScale(d);
70     })
71     .attr("width", xScale.bandwidth());
72
73 ...
```

Task 5.1 D3 Data binding

Home

Lab5.1

Lab5.2

Lab5.3

The D3 journey starts here...

Add

Remove



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lab5-1.html

lab5-3.js

```

lab5 > lab5-3.js > ...
39 d3.select("#add").on("click", function() {
69   .attr("width", xScale.bandwidth())
70   .attr("height", function(d) {
71     return h - yScale(d);
72   });
73 });
74
75 // Add new bars when "Add" button is clicked
76 d3.select("#remove").on("click", function() {
77
78   if (dataset.length > 0) {
79     dataset.pop(); // Remove the last element from the
80
81     // Update xScale domain with the new dataset length
82     xScale.domain(d3.range(dataset.length));
83
84     // Select all rectangles and bind new data
85     var bars = svg.selectAll("rect")
86       .data(dataset);
87
88     // Remove the last bar
89     bars.exit()
90       .transition()
91       .duration(500)
92       .attr("x", w) // Move it off-screen before remove
93       .remove();
94
95     // Update the remaining bars
96     bars.transition()
97       .duration(500)
98       .attr("x", function(d, i) {
99         return xScale(i);
100       })
101       .attr("y", function(d) {
102         return yScale(d);
103       })
104       .attr("width", xScale.bandwidth())
105       .attr("height", function(d) {
106         return h - yScale(d);
107       });
108   }
109 });
110
111

```

Task 5.1 D3 Data binding

Home

Lab5.1

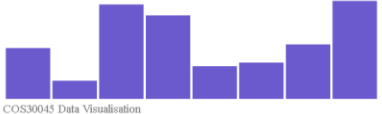
Lab5.2

Lab5.3

The D3 journey starts here...

Add

Remove



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lab5-1.html

lab5-3.js

```

lab5 > lab5-3.js > ...
76 d3.select("#remove").on("click", function() {
94
95   // Update the remaining bars
96   bars.transition()
97     .duration(500)
98     .attr("x", function(d, i) {
99       return xScale(i);
100     })
101     .attr("y", function(d) {
102       return yScale(d);
103     })
104     .attr("width", xScale.bandwidth())
105     .attr("height", function(d) {
106       return h - yScale(d);
107     });
108   }
109 });
110
111

```

Lab6.1

Task 6.1 D3 Data binding

Home

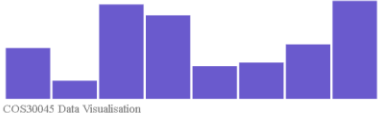
Lab6.1

Lab6.2

The D3 journey starts here...

Add

Remove



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lab6-1.html

home.html

lab6-1.js

```

lab6 > lab6-1.js > ...
1 var w = 500;
2 var h = 130;
3 var barpadding = 2;
4 var maxValue = 25;
5
6 var dataset = [14, 5, 26, 23, 9, 10, 15, 27];
7
8 var xScale = d3.scaleBand()
9   .domain(d3.range(dataset.length))
10   .rangeRound([0, w])
11   .paddingInner(0.05);
12
13 var yScale = d3.scaleLinear()
14   .domain([0, d3.max(dataset)])
15   .range([h, 0]);
16
17 var svg = d3.select("body")
18   .append("svg")
19   .attr("width", w)
20   .attr("height", h);
21
22 // Function to render the bars and bind data
23 function renderBars() {
24   // Select existing bars
25   var bars = svg.selectAll("rect")
26     .data(dataset);
27
28   // Enter new bars
29   var barsEnter = bars.enter()
30     .append("rect")
31     .attr("x", w) // Start off-screen on the right
32     .attr("y", function(d) {
33       return yScale(d);
34     })
35     .attr("width", xScale.bandwidth())
36     .attr("height", function(d) {
37       return h - yScale(d);
38     });
39
40   // Update existing bars
41   bars.transition()
42     .duration(500)
43     .attr("x", function(d, i) {
44       return xScale(i);
45     })
46     .attr("y", function(d) {
47       return yScale(d);
48     })
49     .attr("width", xScale.bandwidth())
50     .attr("height", function(d) {
51       return h - yScale(d);
52     });
53 }
54
55 // Initial render
56 renderBars();
57

```

Task 6.1 D3 Data binding

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HomeLab6 .1Lab6 .2

The D3 journey starts here...

AddRemove



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lab5-1.htmlhome.htmllab6-1.js

lab6 > JS lab6-1.js > ...

```
23 function renderBars() {
24   var barsEnter = bars.enter()
25   .attr("height", function(d) {
26     return h - yScale(d);
27   })
28   .attr("fill", "rgb(106, 90, 205)"); //
29
30   // Merge the enter selection with the update selection
31   barsEnter.merge(bars)
32   .transition()
33   .duration(500)
34   .attr("x", function(d, i) {
35     return xScale(i);
36   })
37   .attr("y", function(d) {
38     return yScale(d);
39   })
40   .attr("width", xScale.bandwidth())
41   .attr("height", function(d) {
42     return h - yScale(d);
43   });
44
45   // Remove exit bars (if there are any)
46   bars.exit()
47   .transition()
48   .duration(500)
49   .attr("x", w)
50   .remove();
51
52   // Remove old text labels
53   svg.selectAll("text").remove();
54
55   // Add event listeners to both new and existing bars
56   barsEnter.merge(bars)
57   .on("mouseover", function(event, d) {
58     d3.select(this)
59       .transition()
60       .duration(200)
61       .attr("fill", "orange");
62
63     var xPos = parseFloat(d3.select(this).attr("x"));
64     var yPos = yScale(d) + 15; // Position above bar
65
66     // Append text when hovering
67     svg.append("text")
68       .attr("id", "tooltip")
69       .attr("x", xPos) // Use the variable d
70       .attr("y", yPos) // Use the variable d
71       .attr("text-anchor", "middle") // Center the text
72       .attr("fill", "black")
73       .text(d);
74
75     .on("mouseout", function() {
76       d3.select("#tooltip").remove();
77       d3.select(this)
78         .transition()
79         .duration(500)
80         .attr("fill", "rgb(106, 90, 205)");
81     });
82   });
83
84   // Initial render
85   renderBars();
86
87   // Add new bars when "Add" button is clicked
88   d3.select("#add").on("click", function() {
89     var newNumber = Math.floor(Math.random() * maxvalue);
90     dataset.push(newNumber);
91
92     // Update xScale domain with the new dataset length
93     xScale.domain(d3.range(dataset.length));
94
95     // Render bars
96     renderBars();
97   });
98
99   // Remove bars when "Remove" button is clicked
100   d3.select("#remove").on("click", function() {
101     if (dataset.length > 0) {
102       dataset.pop(); // Remove the last element from the dataset
103
104       // Update xScale domain with the new dataset length
105       xScale.domain(d3.range(dataset.length));
106
107       // Render bars
108       renderBars();
109     }
110   });
111 }
```

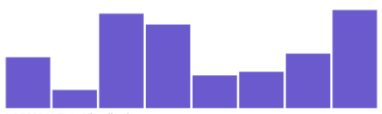
Task 6.1 D3 Data binding

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HomeLab6 .1Lab6 .2

The D3 journey starts here...

AddRemove



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lab5-1.htmlhome.htmllab6-1.js

lab6 > JS lab6-1.js > ...

```
23 function renderBars() {
24   // Add event listeners to both new and existing bars
25   barsEnter.merge(bars)
26     .on("mouseover", function(event, d) {
27       d3.select(this)
28         .transition()
29         .duration(200)
30         .attr("fill", "orange");
31
32       var xPos = parseFloat(d3.select(this).attr("x"));
33       var yPos = yScale(d) + 15; // Position above bar
34
35       // Append text when hovering
36       svg.append("text")
37         .attr("id", "tooltip")
38         .attr("x", xPos) // Use the variable d
39         .attr("y", yPos) // Use the variable d
40         .attr("text-anchor", "middle") // Center the text
41         .attr("fill", "black")
42         .text(d);
43
44       .on("mouseout", function() {
45         d3.select("#tooltip").remove();
46         d3.select(this)
47           .transition()
48           .duration(500)
49           .attr("fill", "rgb(106, 90, 205)");
50       });
51     });
52
53   // Initial render
54   renderBars();
55
56   // Add new bars when "Add" button is clicked
57   d3.select("#add").on("click", function() {
58     var newNumber = Math.floor(Math.random() * maxvalue);
59     dataset.push(newNumber);
60
61     // Update xScale domain with the new dataset length
62     xScale.domain(d3.range(dataset.length));
63
64     // Render bars
65     renderBars();
66   });
67
68   // Remove bars when "Remove" button is clicked
69   d3.select("#remove").on("click", function() {
70     if (dataset.length > 0) {
71       dataset.pop(); // Remove the last element from the dataset
72
73       // Update xScale domain with the new dataset length
74       xScale.domain(d3.range(dataset.length));
75
76       // Render bars
77       renderBars();
78     }
79   });
80 }
```

Task 6.1 D3 Data binding

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HomeLab6 .1Lab6 .2

The D3 journey starts here...

AddRemove



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lab5-1.htmlhome.htmllab6-1.js

lab6 > JS lab6-1.js > ...

```
97 // Add new bars when "Add" button is clicked
98 d3.select("#add").on("click", function() {
99   var newNumber = Math.floor(Math.random() * maxvalue);
100   dataset.push(newNumber);
101
102   // Update xScale domain with the new dataset length
103   xScale.domain(d3.range(dataset.length));
104
105   // Render bars
106   renderBars();
107 });
108
109 // Remove bars when "Remove" button is clicked
110 d3.select("#remove").on("click", function() {
111   if (dataset.length > 0) {
112     dataset.pop(); // Remove the last element from the dataset
113
114     // Update xScale domain with the new dataset length
115     xScale.domain(d3.range(dataset.length));
116
117     // Render bars
118     renderBars();
119   }
120 });
121 }
```

Lab6.2

Task 6.2 D3 Data binding

Home Lab6 .1 Lab6 .2

The D3 journey starts here...

Add Remove Sort



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lab6-1.html lab6-2.js

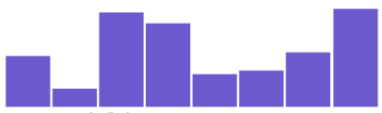
```
lab6 > JS lab6-2.js >
1  var w = 500;
2  var h = 130;
3  var barpadding = 2;
4  var maxValue = 25;
5  var isAscending = true; // Flag to track sort direction
6
7  var dataset = [14, 5, 26, 23, 9, 10, 15, 27];
8
9  var xScale = d3.scaleBand()
10     .domain(d3.range(dataset.length))
11     .rangeRound([0, w])
12     .paddingInner(0.05);
13
14  var yScale = d3.scaleLinear()
15     .domain([0, d3.max(dataset)])
16     .range([h, 0]);
17
18  var svg = d3.select("body")
19     .append("svg")
20     .attr("width", w)
21     .attr("height", h);
22
23  // Function to render the bars and bind data
24  function renderBars() {
25    // Select existing bars
26    var bars = svg.selectAll("rect")
27        .data(dataset);
28
29    // Enter new bars
30    var barsEnter = bars.enter()
31        .append("rect")
32        .attr("x", w) // Start off-screen on the right
33        .attr("y", function(d) {
34            return yScale(d);
35        })
36        .attr("width", xScale.bandwidth());
```

Task 6.2 D3 Data binding

Home Lab6 .1 Lab6 .2

The D3 journey starts here...

Add Remove Sort



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lab6-1.html lab6-2.js

```
lab6 > JS lab6-2.js >
24  function renderBars() {
25    var barsEnter = bars.enter()
26        .attr("y", function(d) {
27            return yScale(d);
28        })
29        .attr("width", xScale.bandwidth())
30        .attr("height", function(d) {
31            return h - yScale(d);
32        })
33        .attr("fill", "rgb(106, 90, 205)"); //
34
35    // Merge the enter selection with the update selection
36    barsEnter.merge(bars)
37        .transition()
38        .duration(500)
39        .attr("x", function(d, i) {
40            return xScale(i);
41        })
42        .attr("y", function(d) {
43            return yScale(d);
44        })
45        .attr("width", xScale.bandwidth())
46        .attr("height", function(d) {
47            return h - yScale(d);
48        });
49
50    // Remove exit bars (if there are any)
51    bars.exit()
52        .transition()
53        .duration(500)
54        .attr("x", w)
55        .remove();
56
57    // Remove old text labels
58    svg.selectAll("text").remove();
59  }
```


Task 6.2 D3 Data binding

Home

Lab6.1

Lab6.2

The D3 journey starts here...

Add

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lab5-1.html

lab6-2.js

lab6 > lab6-2.js > ...

```
function renderBars() {  
  // Remove old text labels  
  svg.selectAll("text").remove();  
  
  // Add event listeners to both new and existing bars  
  barsEnter.merge(bars)  
    .on("mouseover", function(event, d) {  
      d3.select(this)  
        .transition()  
        .duration(200)  
        .attr("fill", "orange");  
  
      var xPositon = parseFloat(d3.select(this).attr("x"));  
      var yPositon = yScale(d) + 15; // Position above the bar  
  
      // Append text when hovering  
      svg.append("text")  
        .attr("id", "tooltip")  
        .attr("x", xPositon) // Use the variable d  
        .attr("y", yPositon) // Use the variable d  
        .attr("text-anchor", "middle") // Center the text  
        .attr("fill", "black")  
        .text(d);  
    })  
    .on("mouseout", function() {  
      d3.select("#tooltip").remove();  
      d3.select(this)  
        .transition()  
        .duration(500)  
        .attr("fill", "rgb(106, 90, 205)");  
    });  
}  
  
// Initial render  
renderBars();
```

Task 6.2 D3 Data binding

Home

Lab6.1

Lab6.2

The D3 journey starts here...

Add

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Sort



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lab5-1.html

lab6-2.js

lab6 > lab6-2.js > ...

```
function renderBars() {  
  // Initial render  
  renderBars();  
  
  // Add new bars when "Add" button is clicked  
  d3.select("#add").on("click", function() {  
    var newNumber = Math.floor(Math.random() * maxValue);  
    dataset.push(newNumber);  
  
    // Update xScale domain with the new dataset length  
    xScale.domain(d3.range(dataset.length));  
  
    // Render bars  
    renderBars();  
  });  
  
  // Remove bars when "Remove" button is clicked  
  d3.select("#remove").on("click", function() {  
    if (dataset.length > 0) {  
      dataset.pop(); // Remove the last element from the dataset  
  
      // Update xScale domain with the new dataset length  
      xScale.domain(d3.range(dataset.length));  
  
      // Render bars  
      renderBars();  
    }  
  });  
  
  // Sort bars when "Sort" button is clicked  
  d3.select("#sort").on("click", function() {  
    dataset.sort(function(a, b) {  
      return isAscending ? d3.ascending(a, b) : d3.descending(a, b);  
    });  
  
    // Toggle the sort direction for next click  
    isAscending = !isAscending;  
  
    // Update the xScale domain  
    xScale.domain(d3.range(dataset.length));  
  
    // Re-render bars after sorting  
    renderBars();  
  });  
}
```

Task 6.2 D3 Data binding

Home

Lab6.1

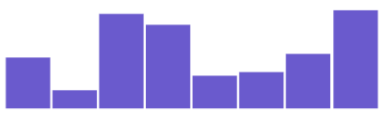
Lab6.2

The D3 journey starts here...

Add

Remove

Sort



COS30045 Data Visualisation
Lim Chi Yuen

lab5-1.html

lab6-2.js

lab6 > lab6-2.js > ...

```
function renderBars() {  
  // Initial render  
  renderBars();  
  
  // Add new bars when "Add" button is clicked  
  d3.select("#add").on("click", function() {  
    var newNumber = Math.floor(Math.random() * maxValue);  
    dataset.push(newNumber);  
  
    // Update xScale domain with the new dataset length  
    xScale.domain(d3.range(dataset.length));  
  
    // Render bars  
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  // Remove bars when "Remove" button is clicked  
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      xScale.domain(d3.range(dataset.length));  
  
      // Render bars  
      renderBars();  
    }  
  });  
  
  // Sort bars when "Sort" button is clicked  
  d3.select("#sort").on("click", function() {  
    dataset.sort(function(a, b) {  
      return isAscending ? d3.ascending(a, b) : d3.descending(a, b);  
    });  
  
    // Toggle the sort direction for next click  
    isAscending = !isAscending;  
  
    // Update the xScale domain  
    xScale.domain(d3.range(dataset.length));  
  
    // Re-render bars after sorting  
    renderBars();  
  });  
}
```